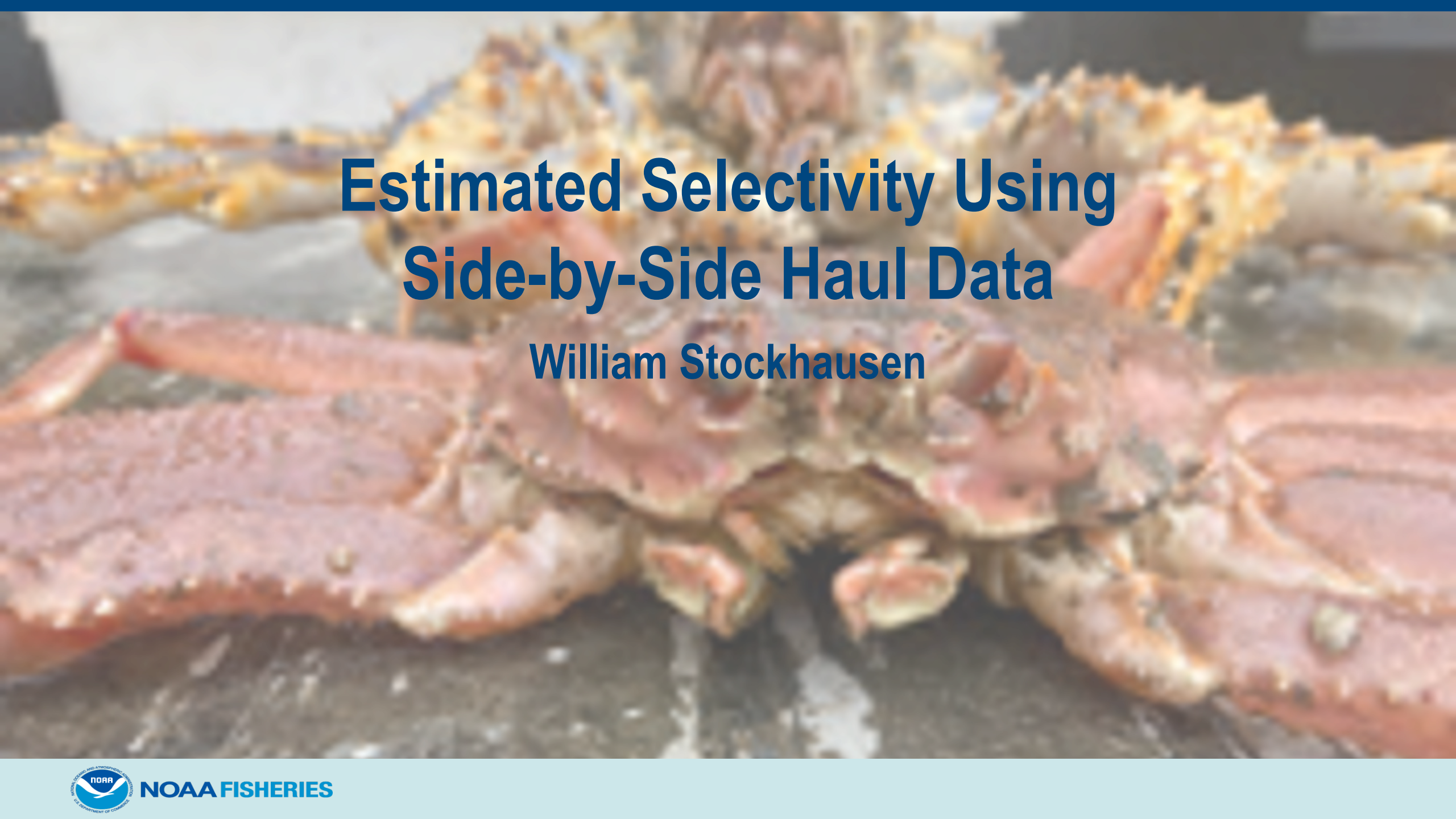




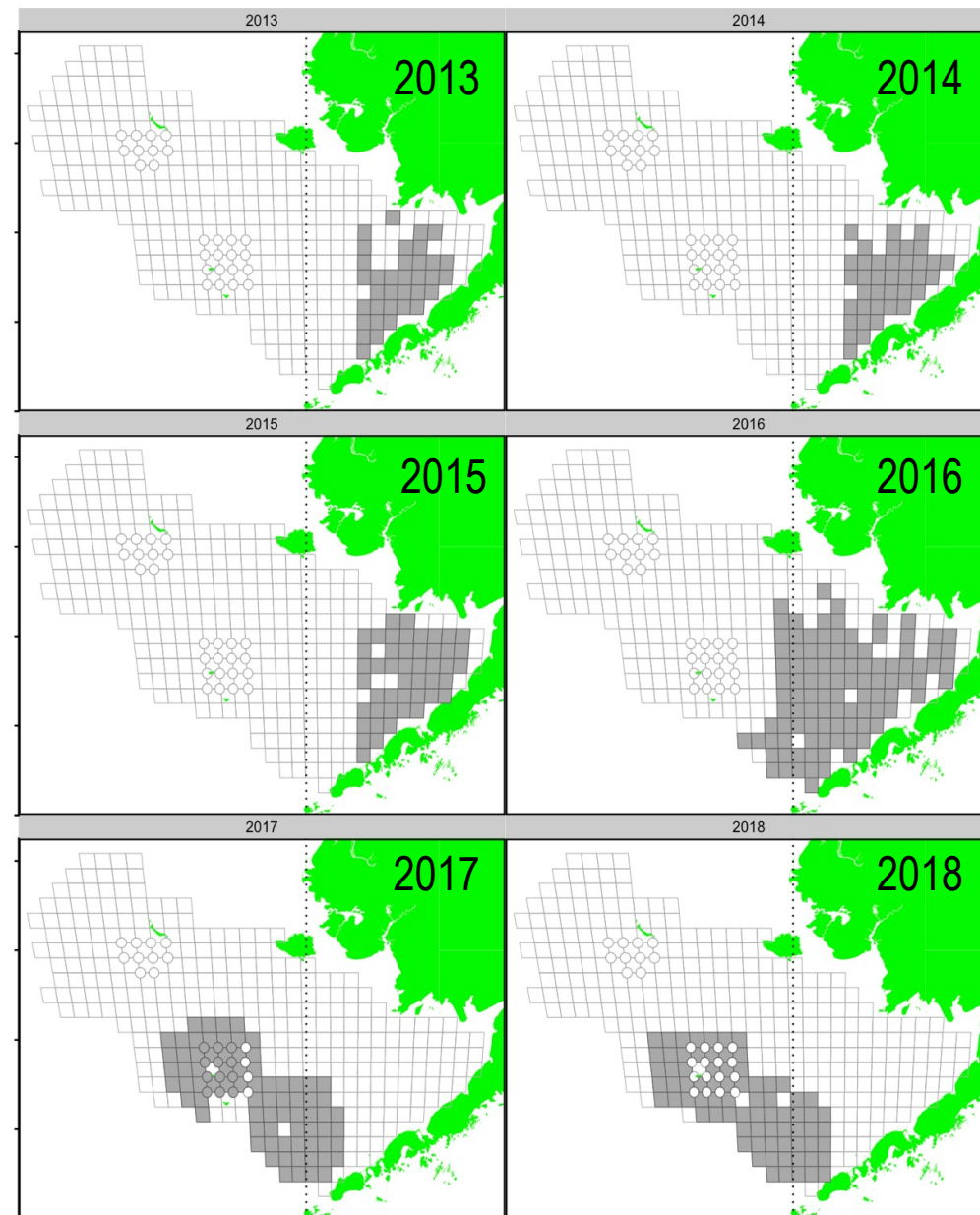
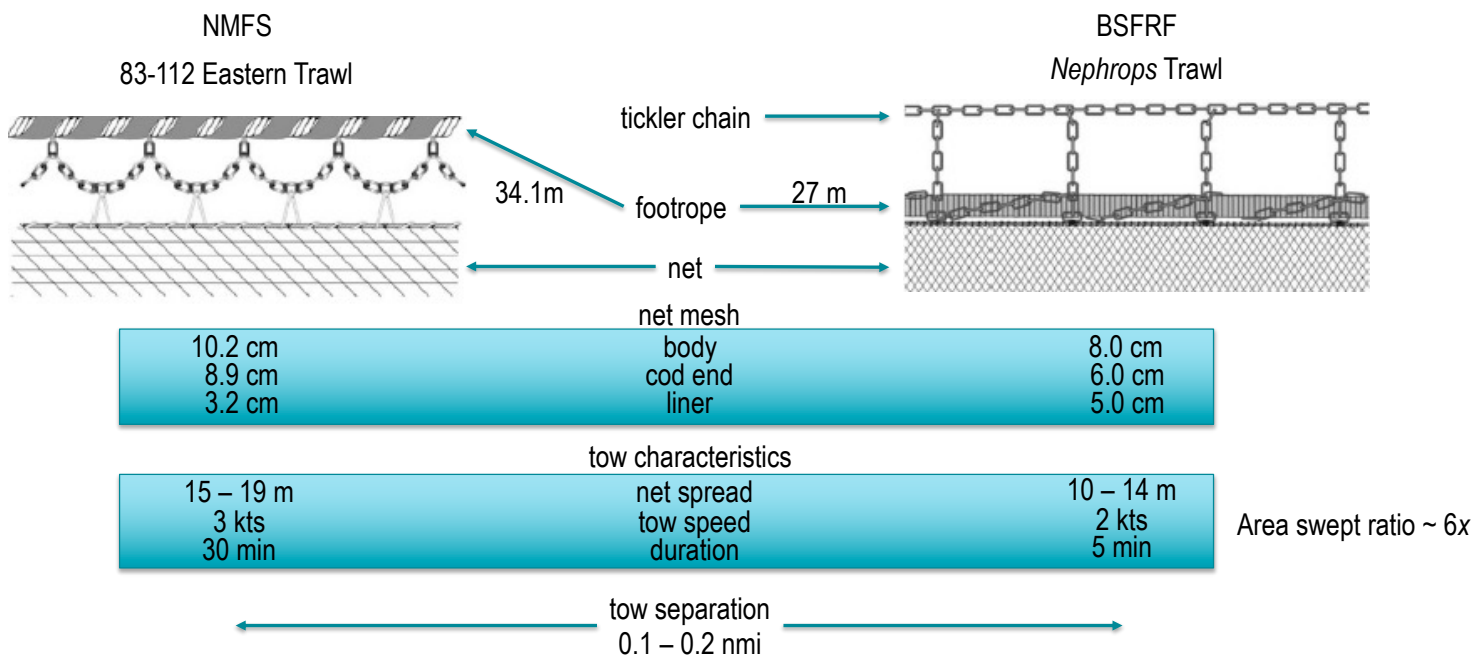
Estimated Selectivity Using Side-by-Side Haul Data

William Stockhausen



NOAA FISHERIES

BSFRF-NMFS “Side-By-Side” Selectivity Studies



The Math

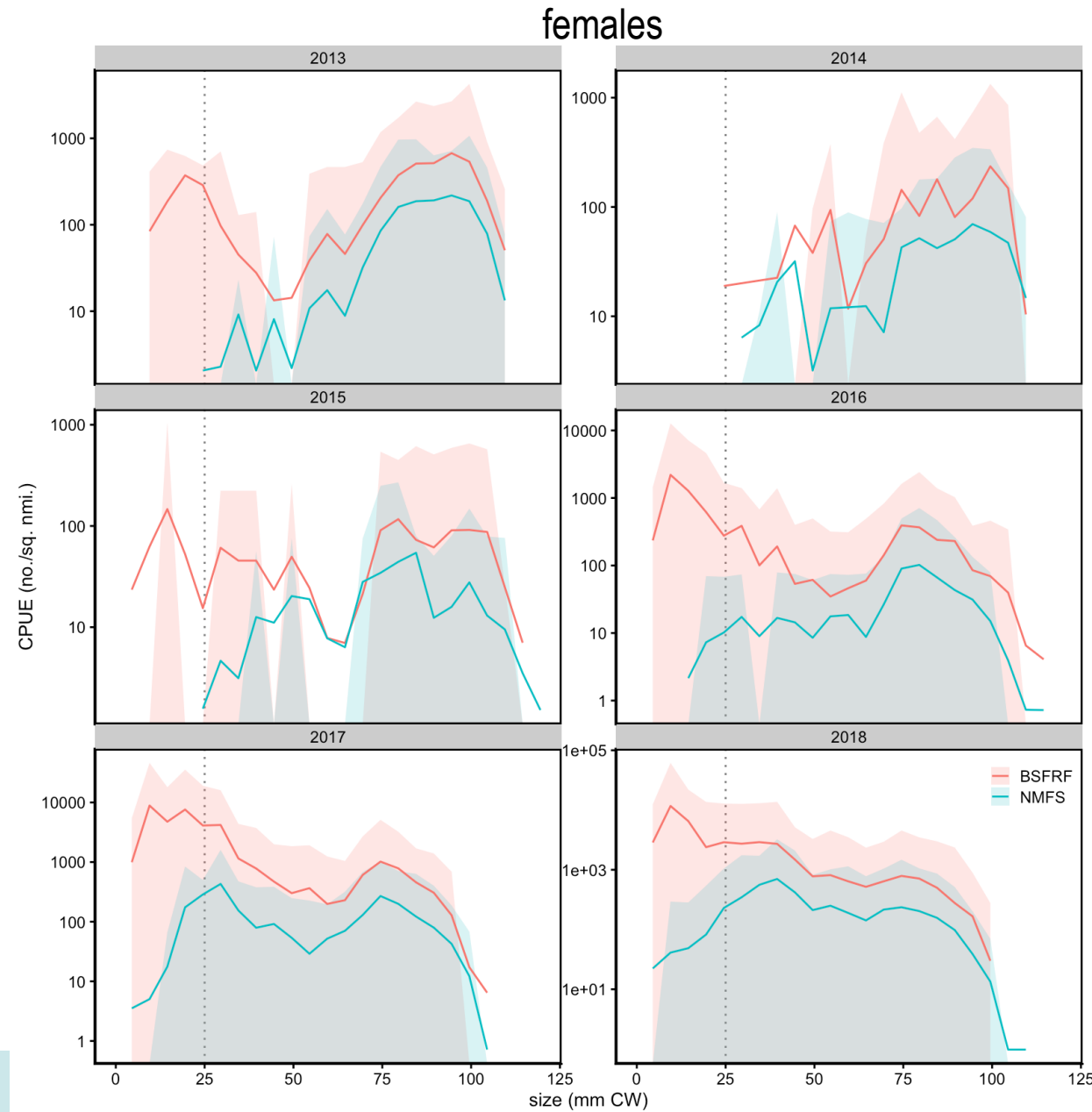
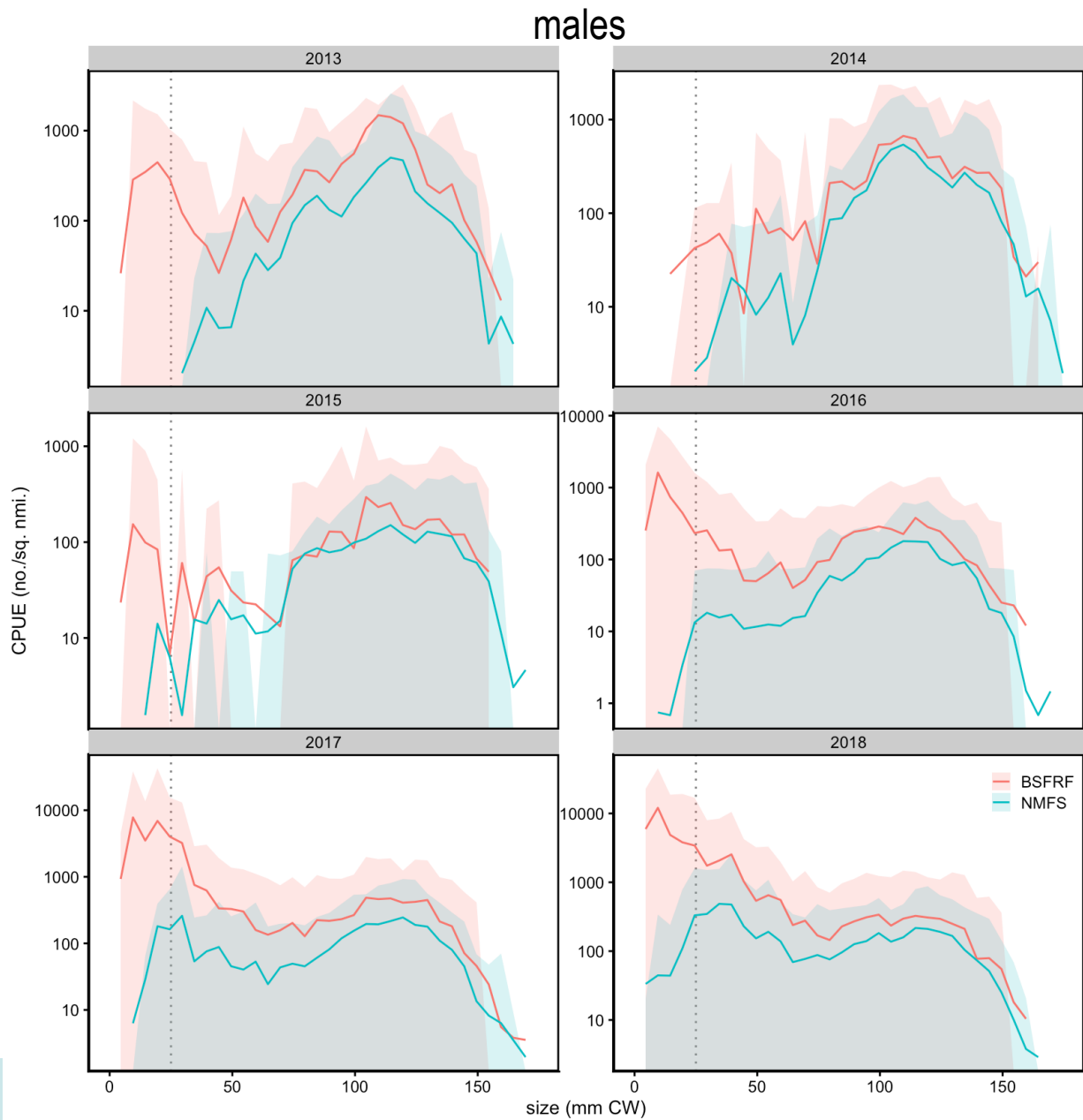
- c : gear-specific CPUE (number caught/unit area)
- q : gear-specific catchability/selectivity
- d : crab density (number/unit area)
- z : size (sex, maturity state, shell condition, etc.)

$$\begin{array}{l} c_z^N \equiv q_z^N \cdot d_z \\ c_z^B \equiv q_z^B \cdot d_z \end{array} \quad \longrightarrow \quad \frac{c_z^N}{c_z^B} = \frac{q_z^N \cdot d_z}{q_z^B \cdot d_z} = \frac{q_z^N}{q_z^B} = R_z \equiv \text{relative catchability}$$

- **BIG** assumption 1: $q_z^B = 1 \longrightarrow R_z = \frac{c_z^N}{c_z^B} = q_z^N \equiv \text{absolute catchability}$
- **BIG** assumption 2: d_z is (statistically) the same for both gears
- Applications
 - haul level (location-specific) $c_z^{\text{haul}} = q_z^{\text{haul}} \cdot d_z^{\text{station area}}$
 - survey-level (area-aggregated) $c_z^{\text{survey}} = q_z^{\text{survey}} \cdot d_z^{\text{survey area}}$

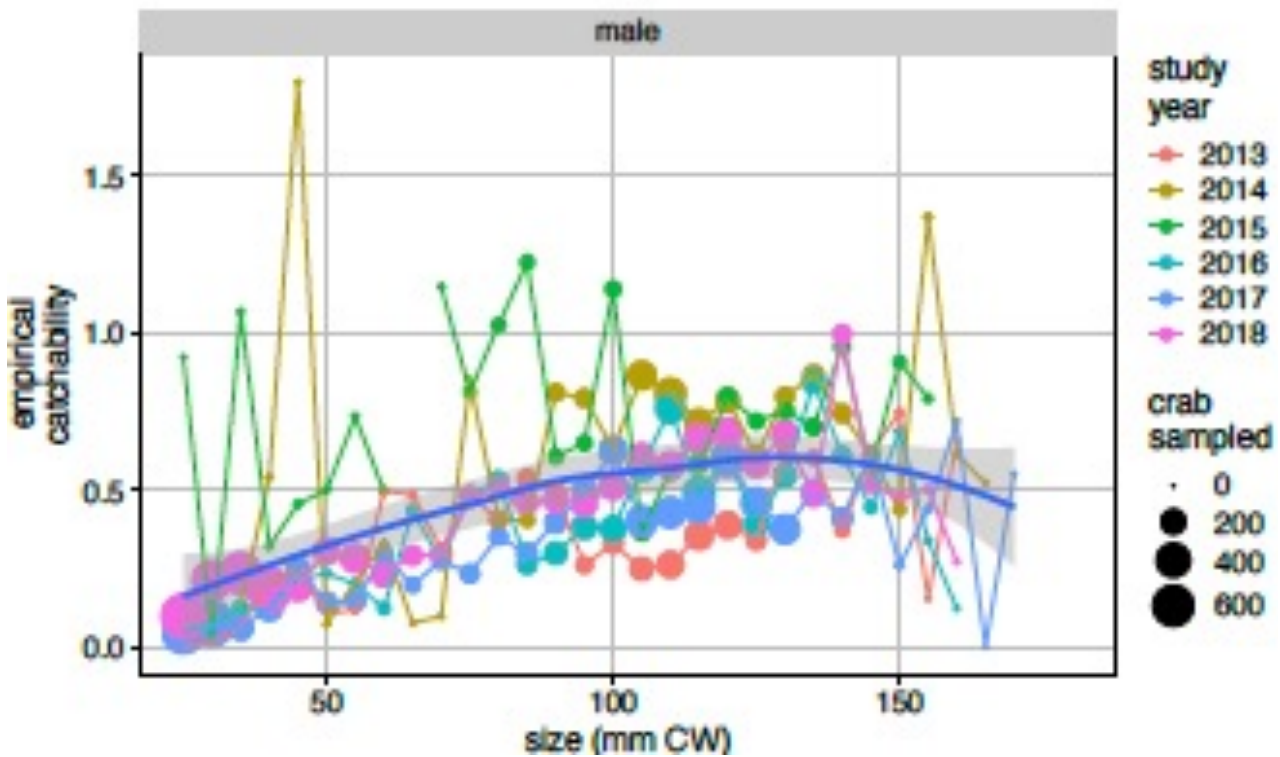
Survey Level (Aggregated) Analysis

Area-expanded Bootstrapped Size Compositions: Tanner Crab

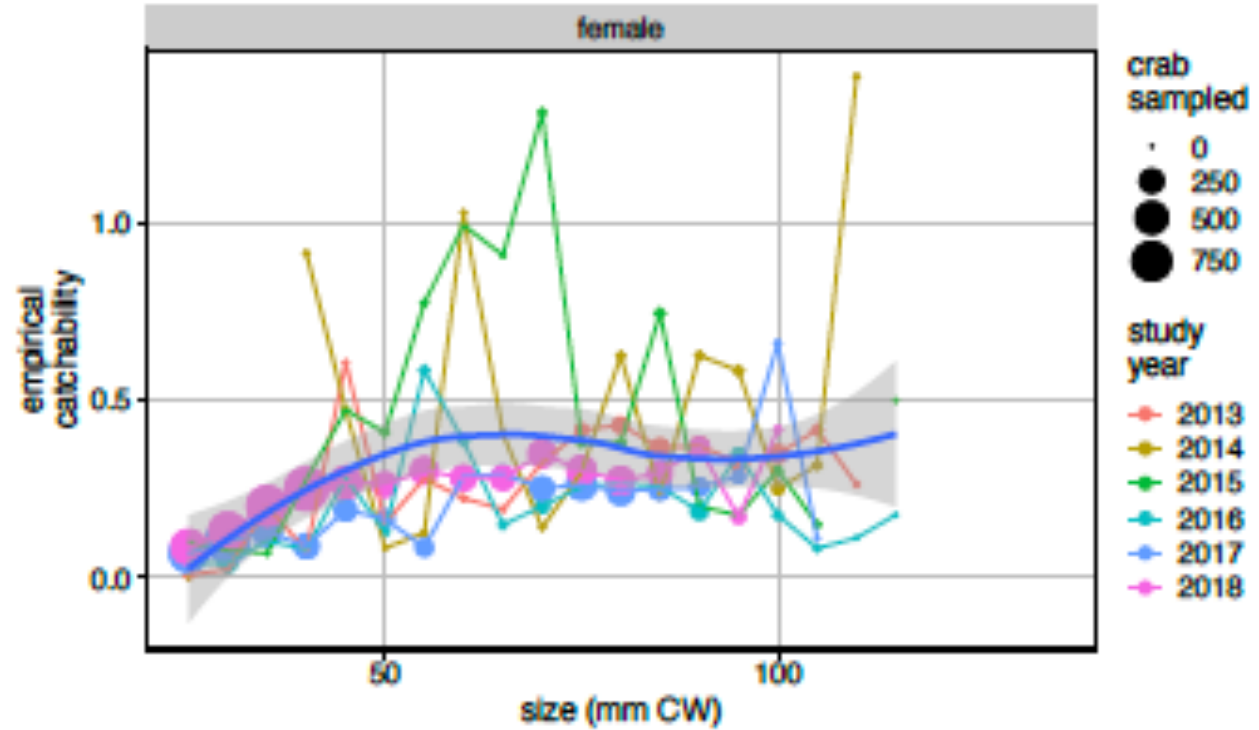


Empirical Survey-level Catchability Curves

males



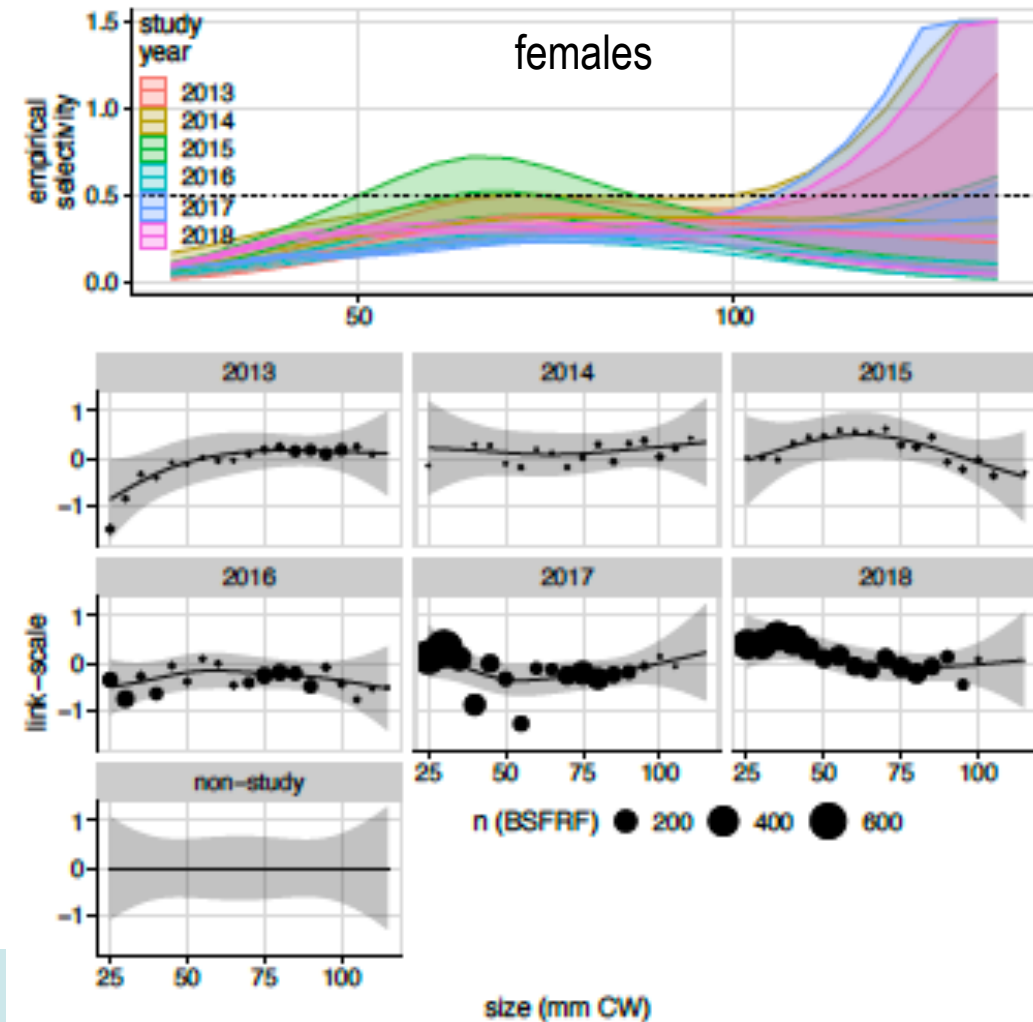
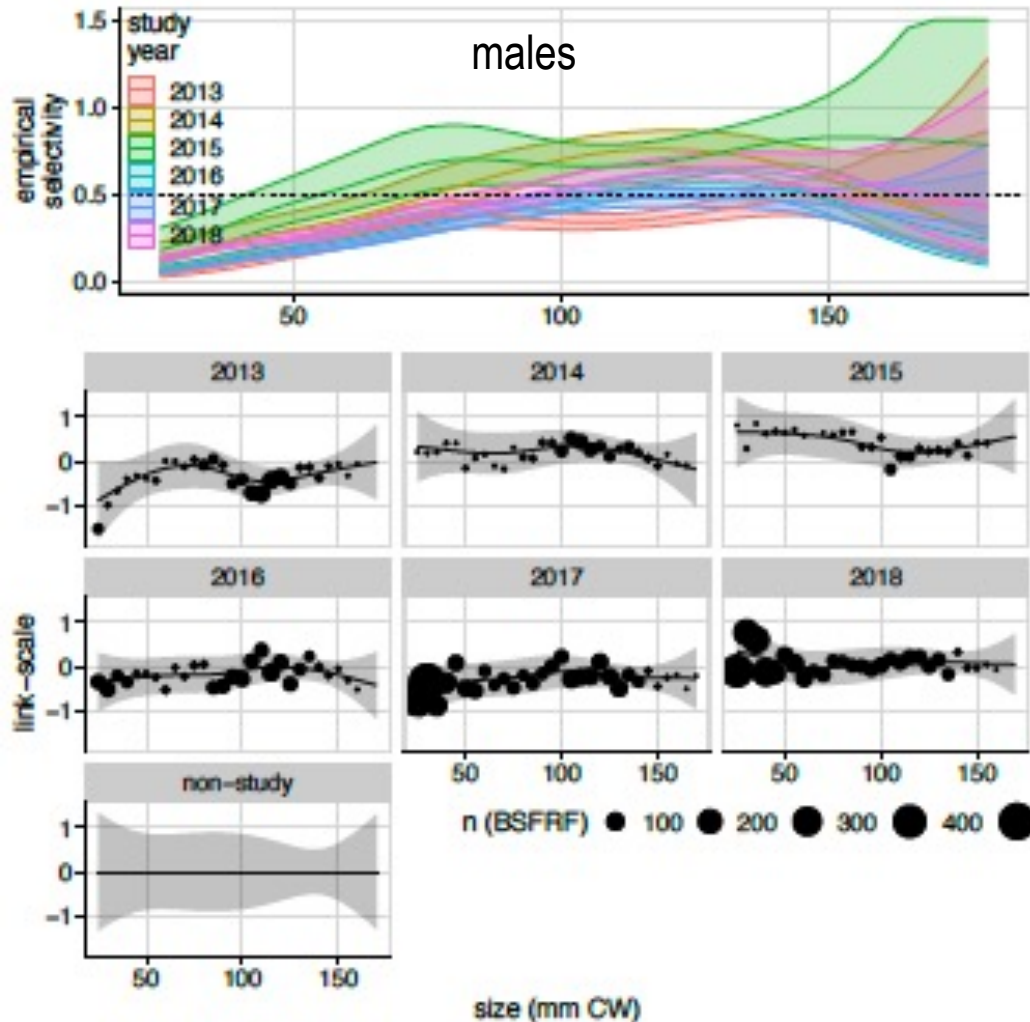
females



Estimated Catchability: Best Survey-Level (Area-aggregated) Models

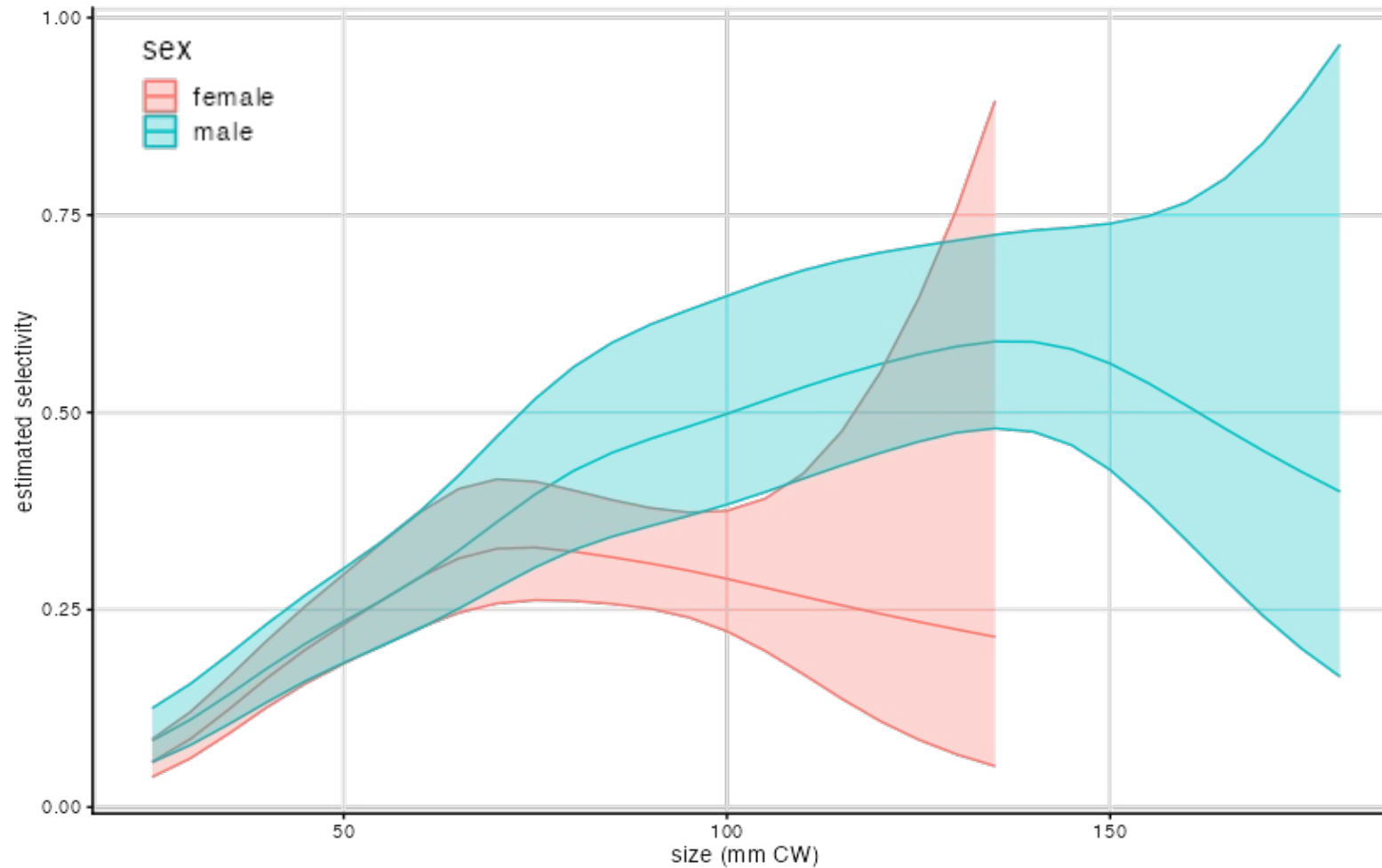
- model 1: $R \sim s(z, bs="tp", k=10)$
- model 2: $R \sim s(z, bs="tp", k=10) + ti(z, by=y, bs="sz")$ with year as fixed effect
- model 3: $R \sim s(z, bs="tp", k=10) + s(z, y, bs="fs")$ with year as random effect

all models:
family: tweedie (log link)
weights: $n_z^B / \text{mean}(n_z^B)$



Estimated Catchability: Best Survey-Level (Area-aggregated) Models

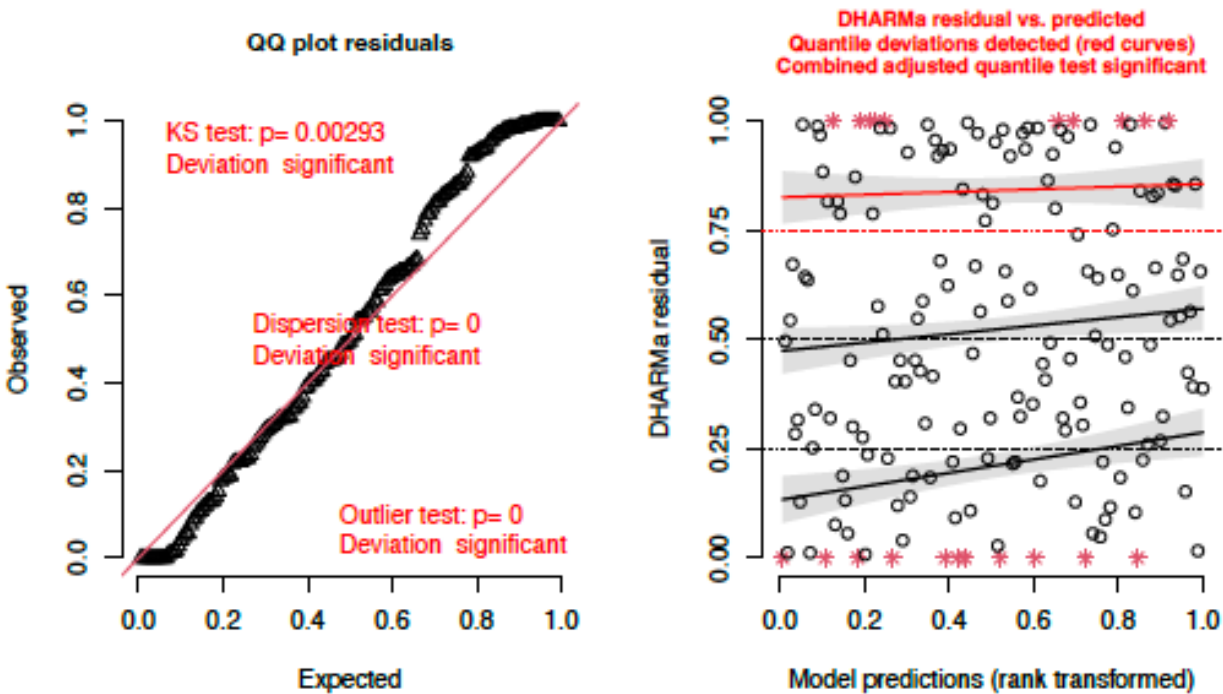
- best models: model 3 for both males and females



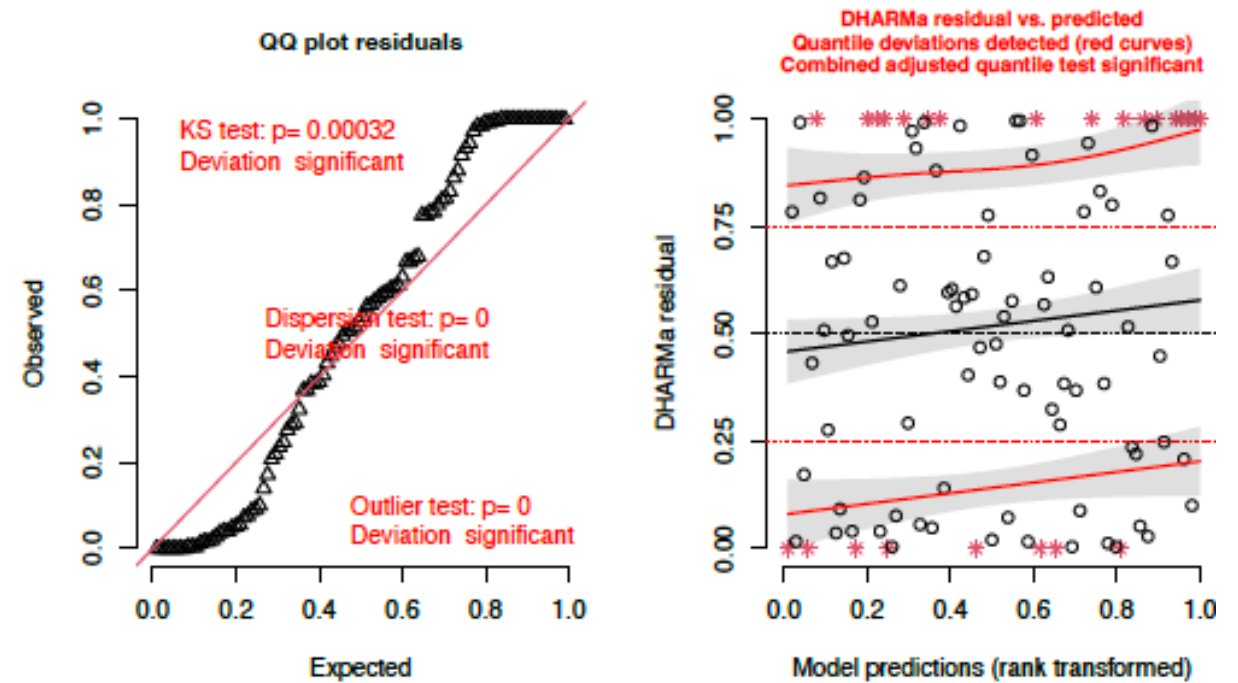
Best Survey-Level (Area-aggregated) Models

- best models: model 3 for both males and females

males



females



Haul Level Analysis

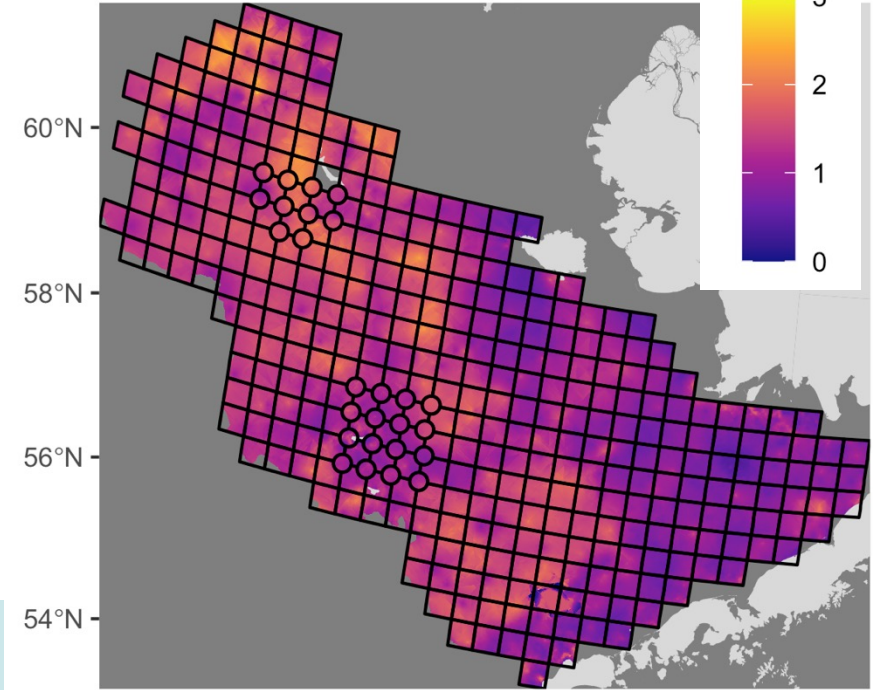
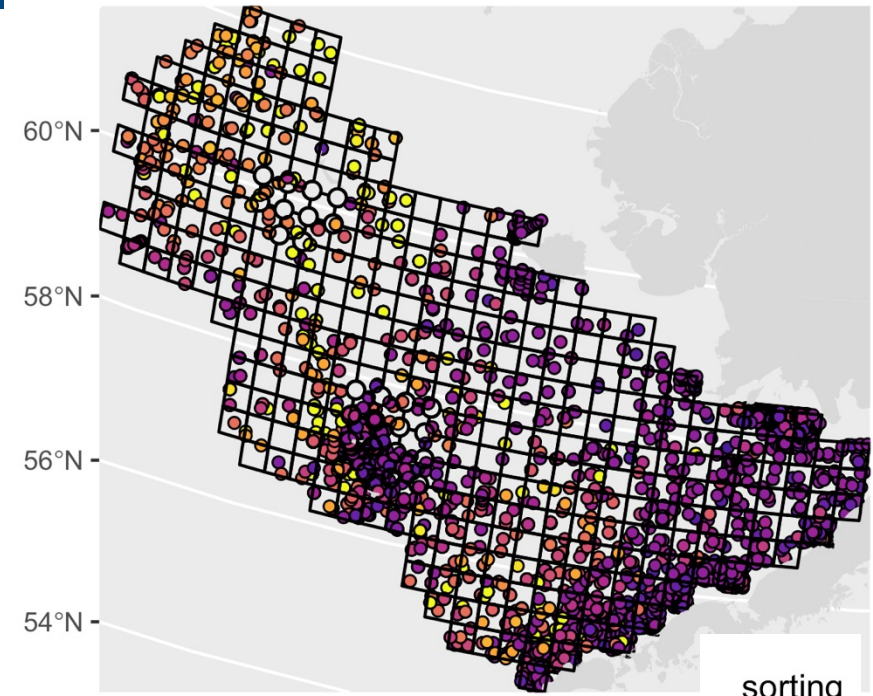
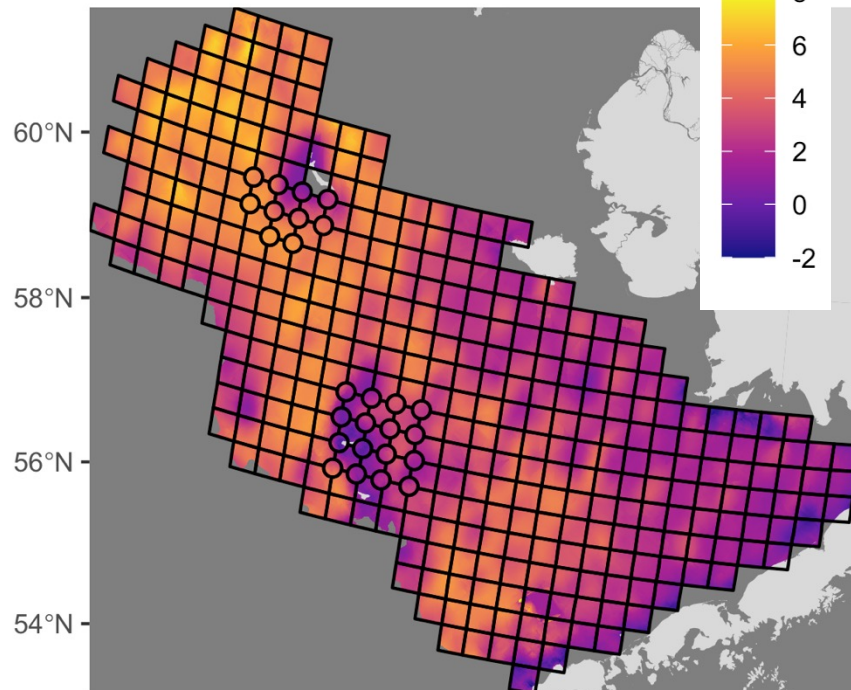
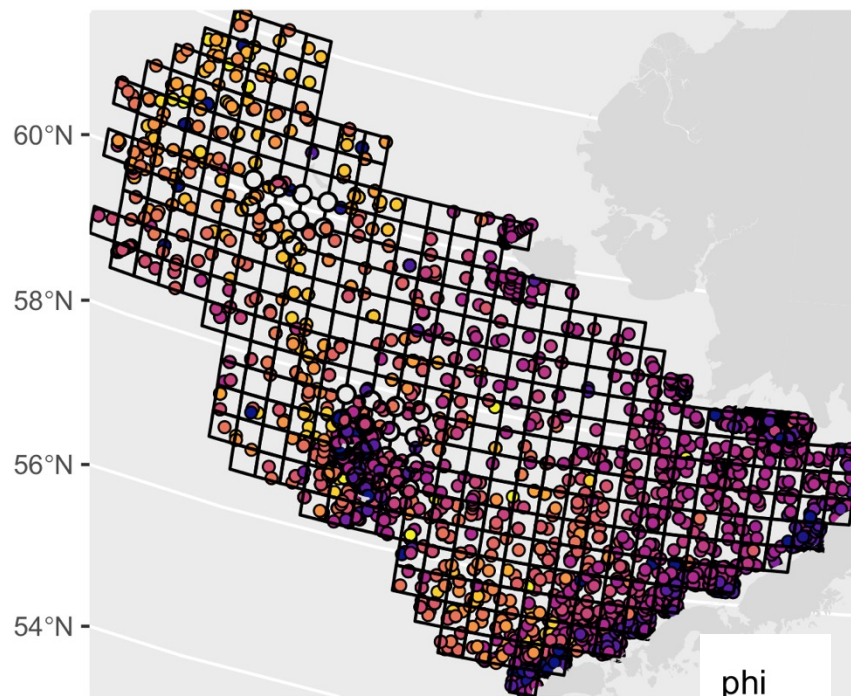
Potential environmental covariates

ϕ (f): measure of
In-scale mean grain size

sorting (s): measure of
In-scale grain size variance

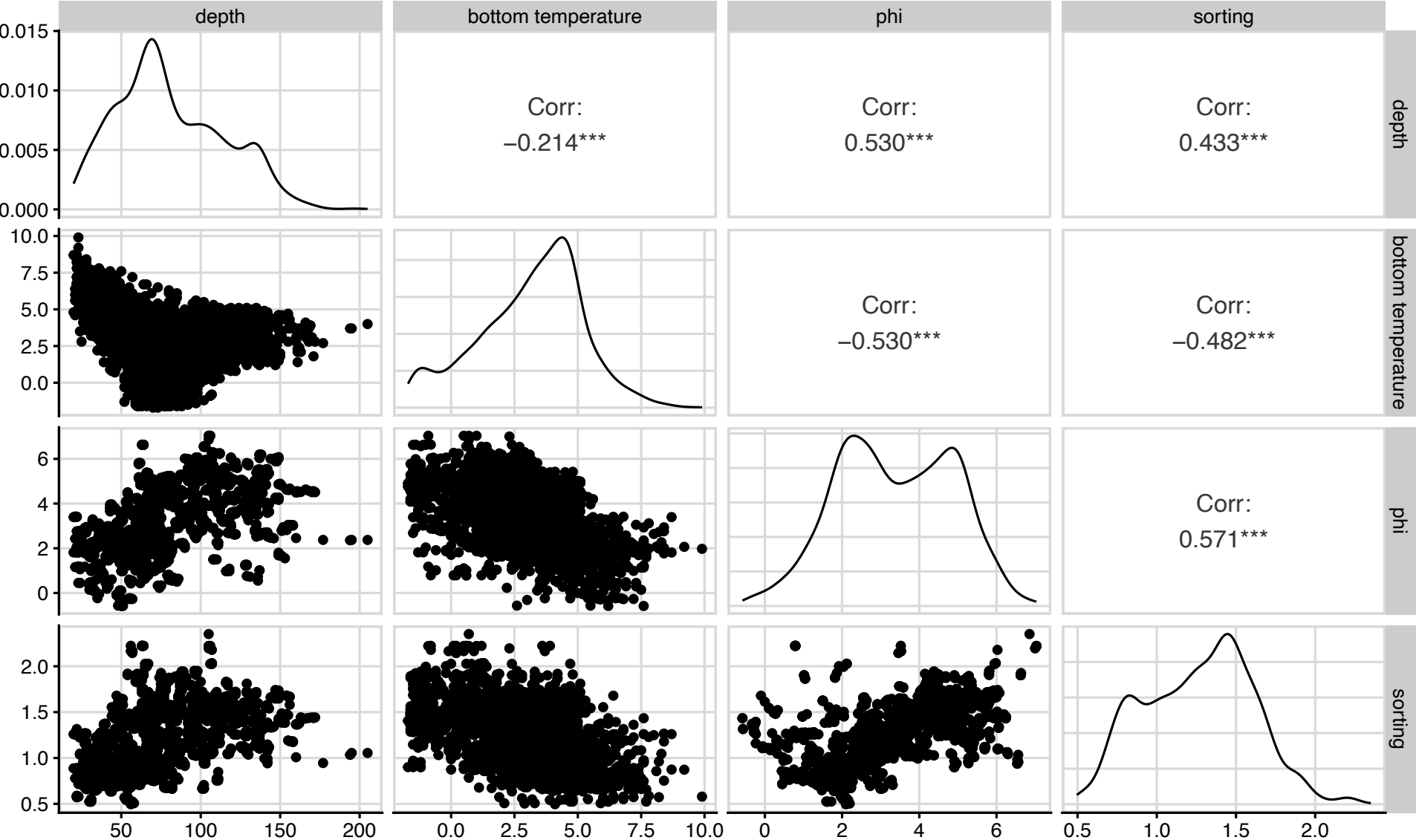
depth (d)

bottom temperature (t)



Potential haul-specific environmental covariates

based on NMFS haul



bottom depth (d)

bottom temp. (t)

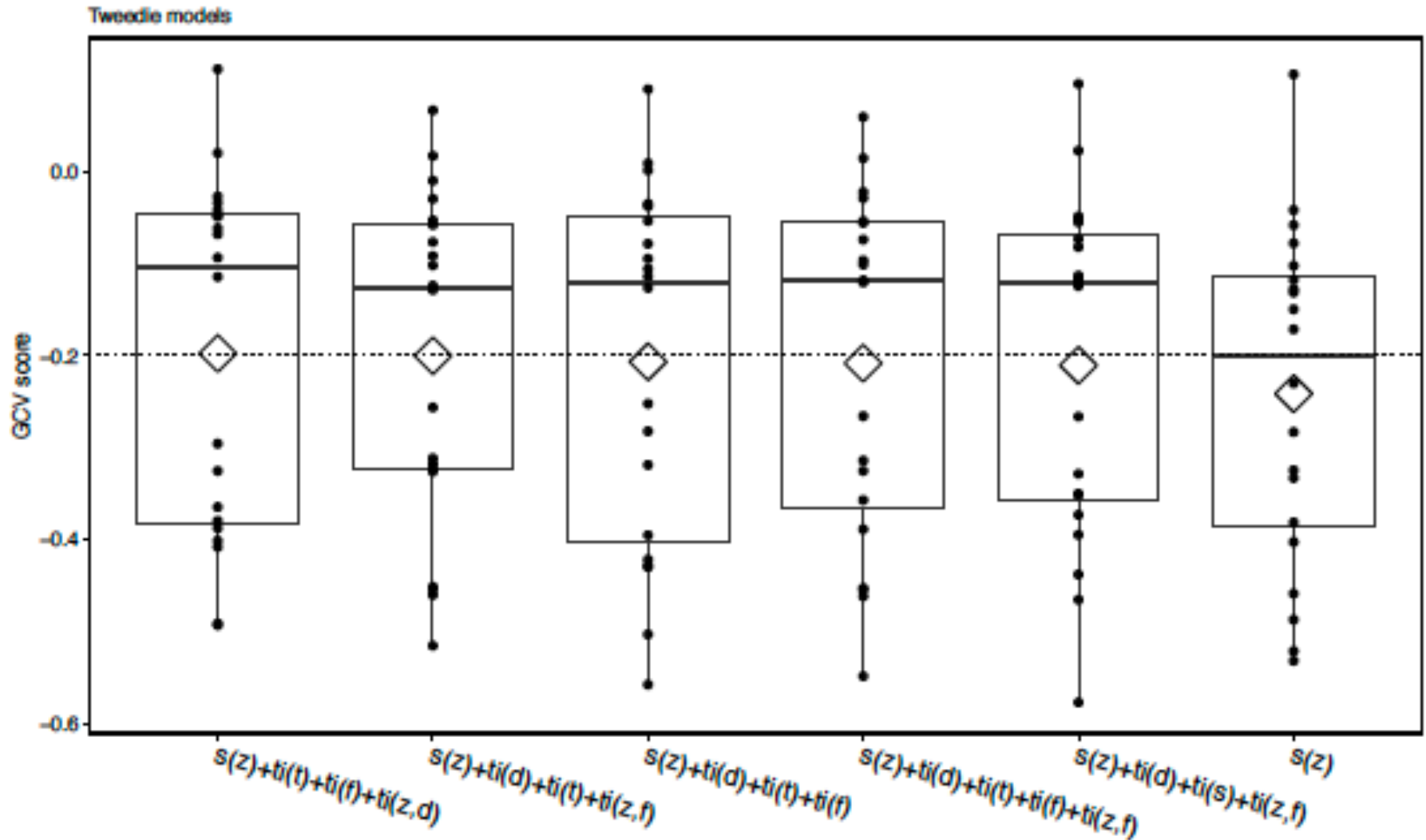
phi (f): mean ln-scale grain size

sorting coefficient (s): measure of ln-scale grain size variance

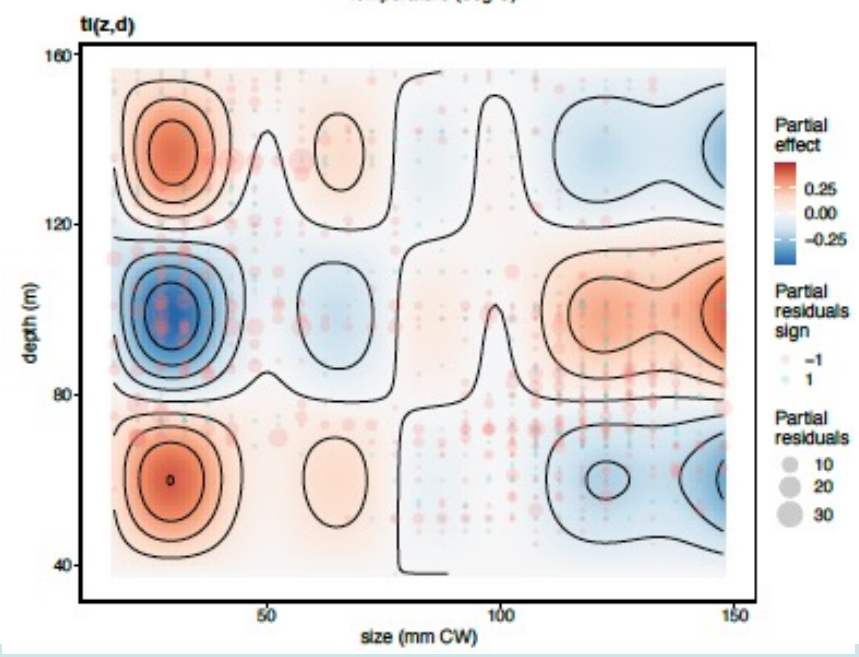
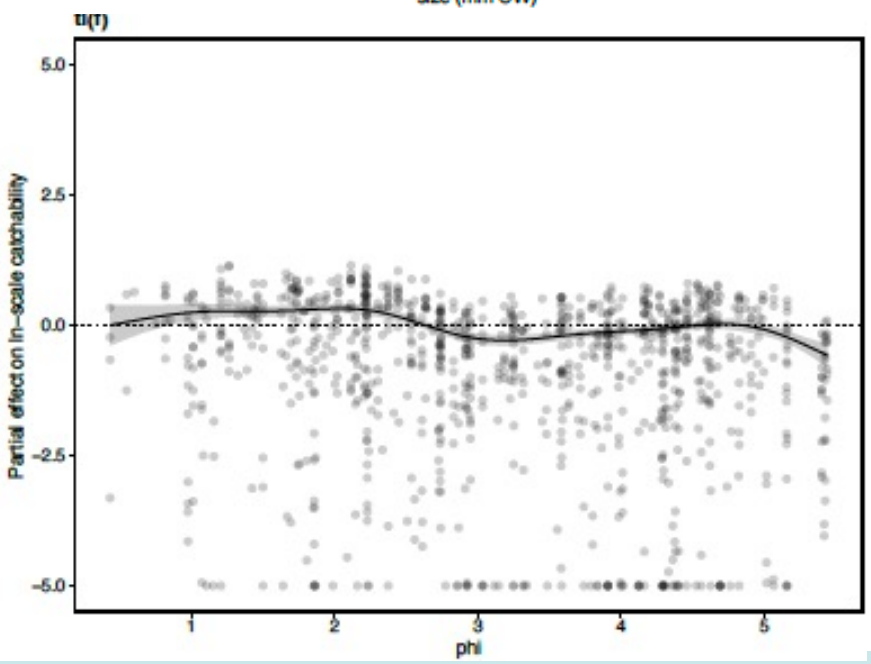
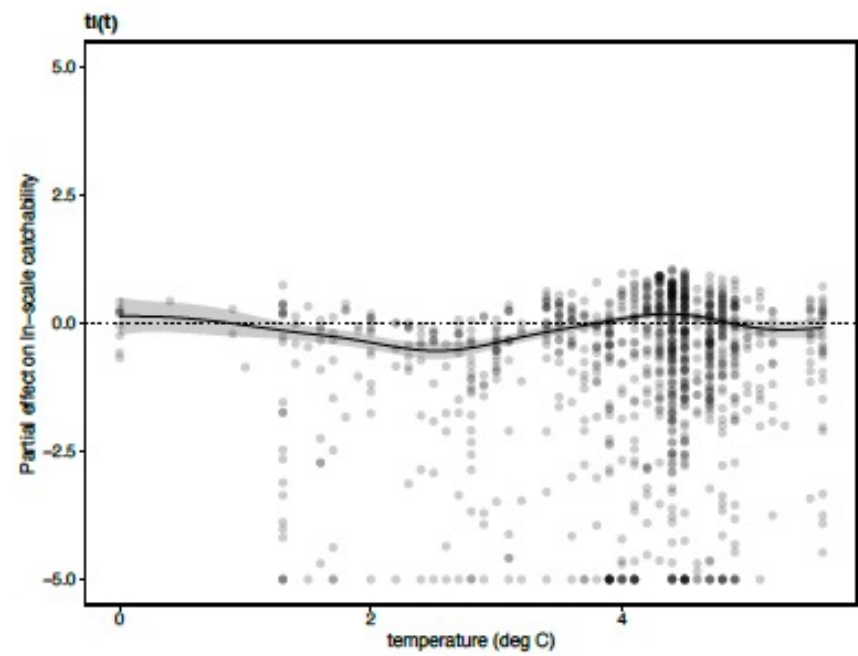
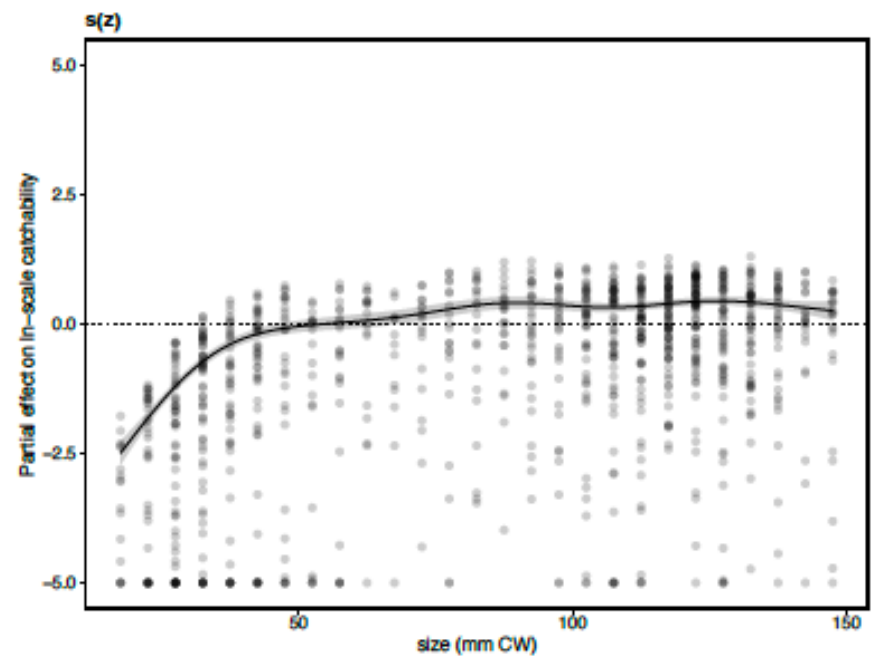
$R \sim s(z, bs="ts", k=k1) +$
 $ti(d, bs="ts", k=k2) + ti(t, bs="ts", k=k2) + ti(f, bs="ts", k=k2) + ti(s, bs="ts", k=k2) +$
 $ti(z, d, bs="ts", k=c(k1, k2)) + ti(z, t, bs="ts", k=c(k1, k2)) + ti(z, f, bs="ts", k=c(k1, k2)) + ti(z, s, bs="ts", k=c(k1, k2))$
 with $tw(link="log")$

Model selection: GCV

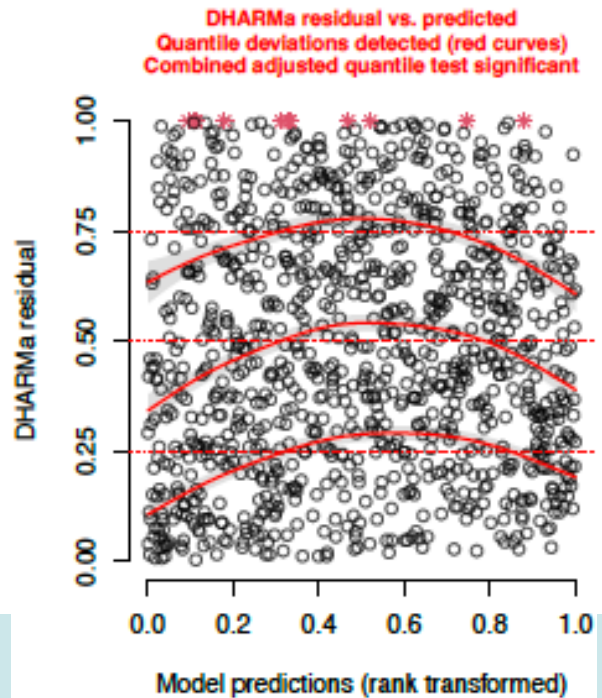
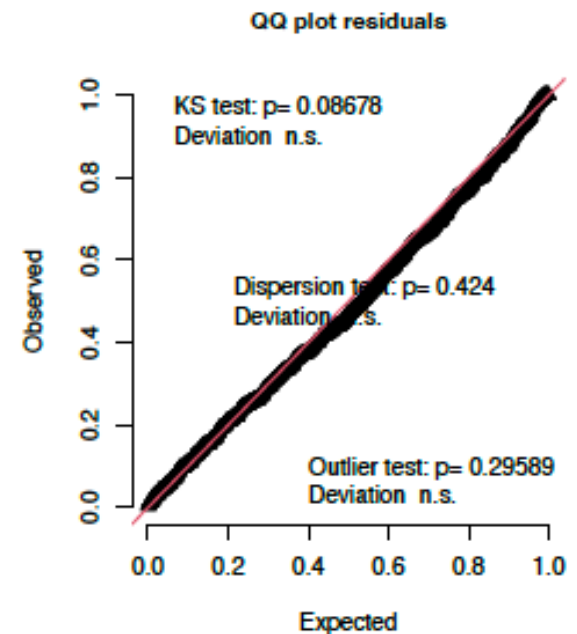
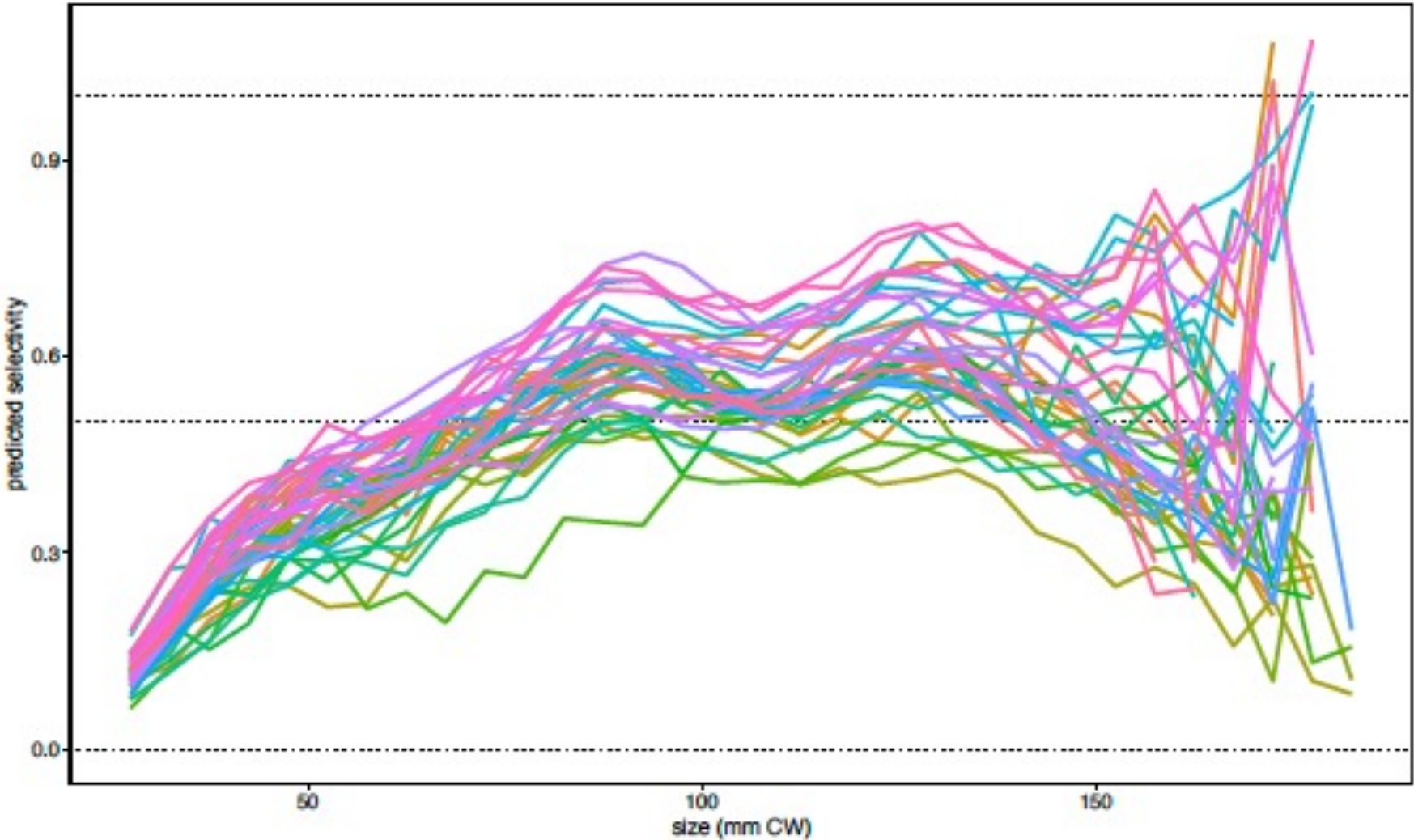
- 256 candidates
- 20 folds
- Best GCV score



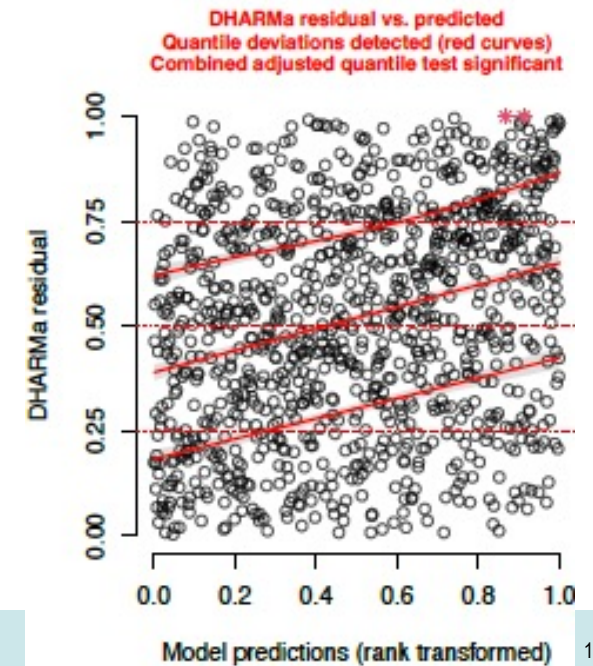
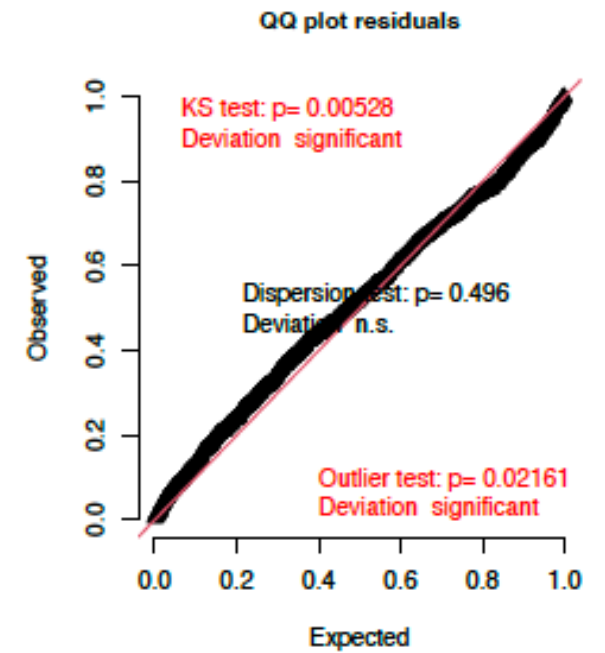
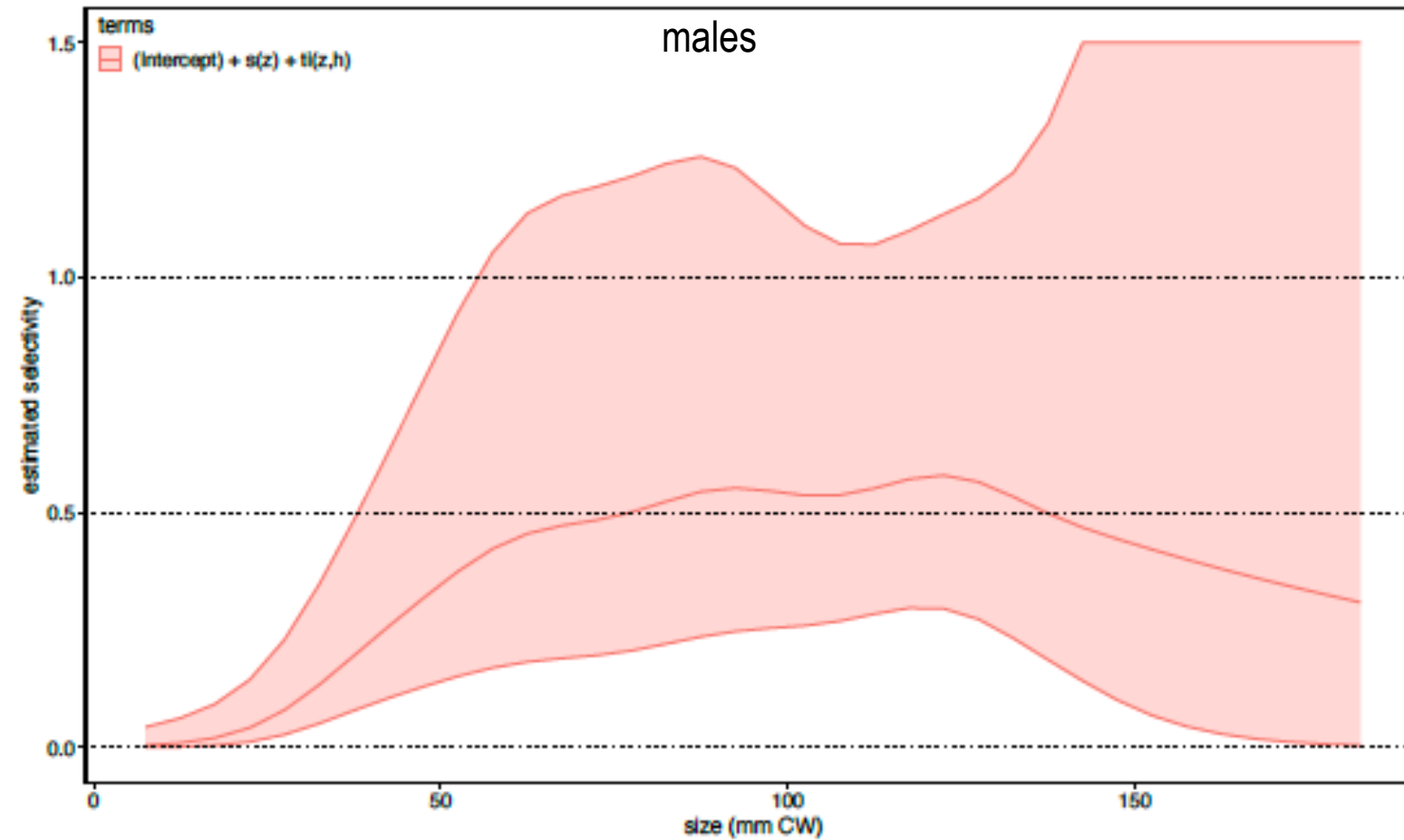
Marginal effects



CPUE-averaged annual survey catchability



$R \sim s(z, bs="ts", k=k1) + ti(z, h, bs="fs", k=k1)$
 with $tw(link="log");$



Thygesen et al Approach

Thygesen, U. H., Kristensen, K., Jansen, T., and Beyer, J. E. Intercalibration of survey methods using paired fishing operations and log-Gaussian Cox processes. – ICES Journal of Marine Science, 76: 1189–1199.

Ratio R modeled as a Log-Gaussian Cox process

- $N_{ijk} | \Phi, R, S \sim \text{Poisson}(A_{ij} \cdot \exp(S_{jk} + \Phi_{ik} + R_{ijk}))$ numbers at station i by gear j in size bin k
- Φ_{ik} characterizes the **ln-scale** size distribution at station i , size bin k
- R_{ijk} characterizes **ln-scale** haul/gear-specific variations in size distribution at station i
- S_{jk} characterizes the gear-specific **ln-scale** selectivity for size bin k
- A_{ij} is the area swept by gear j at station i (known)
- No environmental covariates included in this formalism
- BSFRF: gear = 2. BSFRF selectivity = 1 $\Rightarrow S_{2k} = 0$ (ln-scale!)
- in absence of R_{ijk} ,

$$\exp(S_{1k}) = \frac{\mathbb{E}\{N_{i1k}/A_{i1} | \Phi, S\}}{\mathbb{E}\{N_{i2k}/A_{i1} | \Phi\}}$$

Statistical Modeling

- Statistical assumptions

$S_{1(k+1)} - S_{1k} \sim N(0, \sigma_S^2)$ for $k = 1, \dots, m_1 - 1$ In-scale change is RW in size

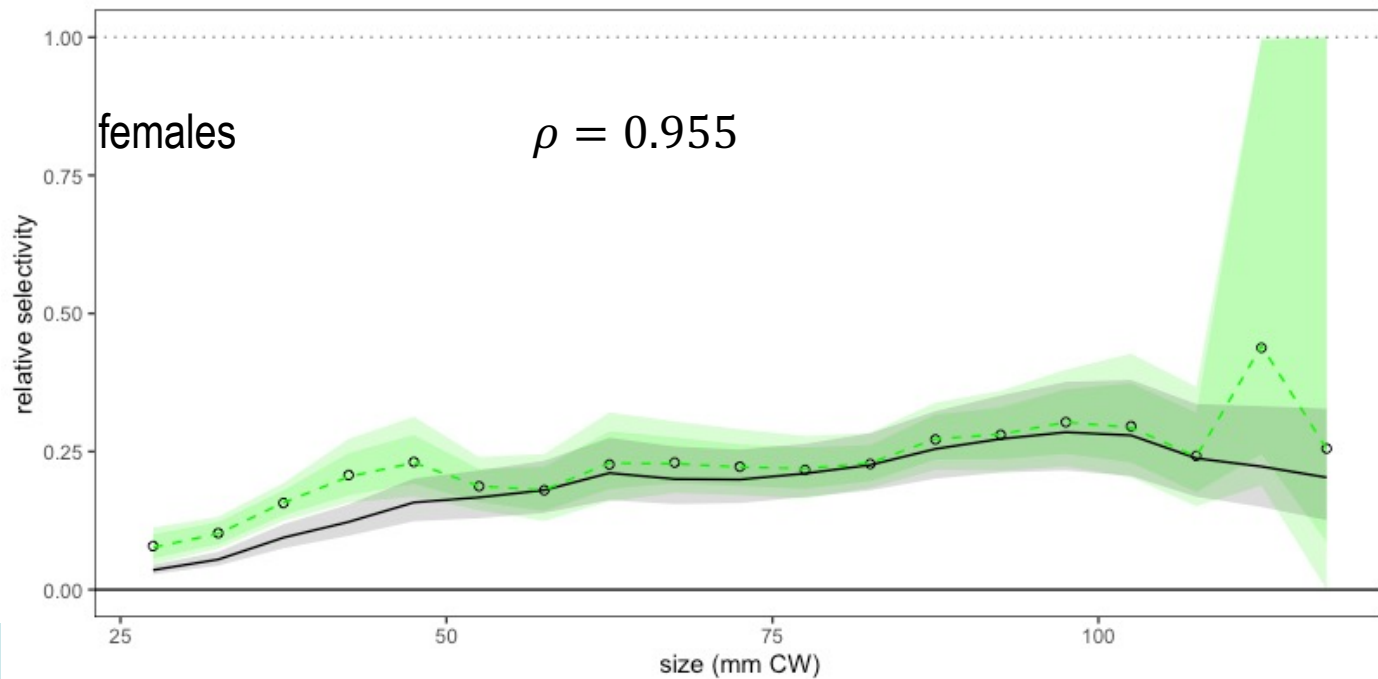
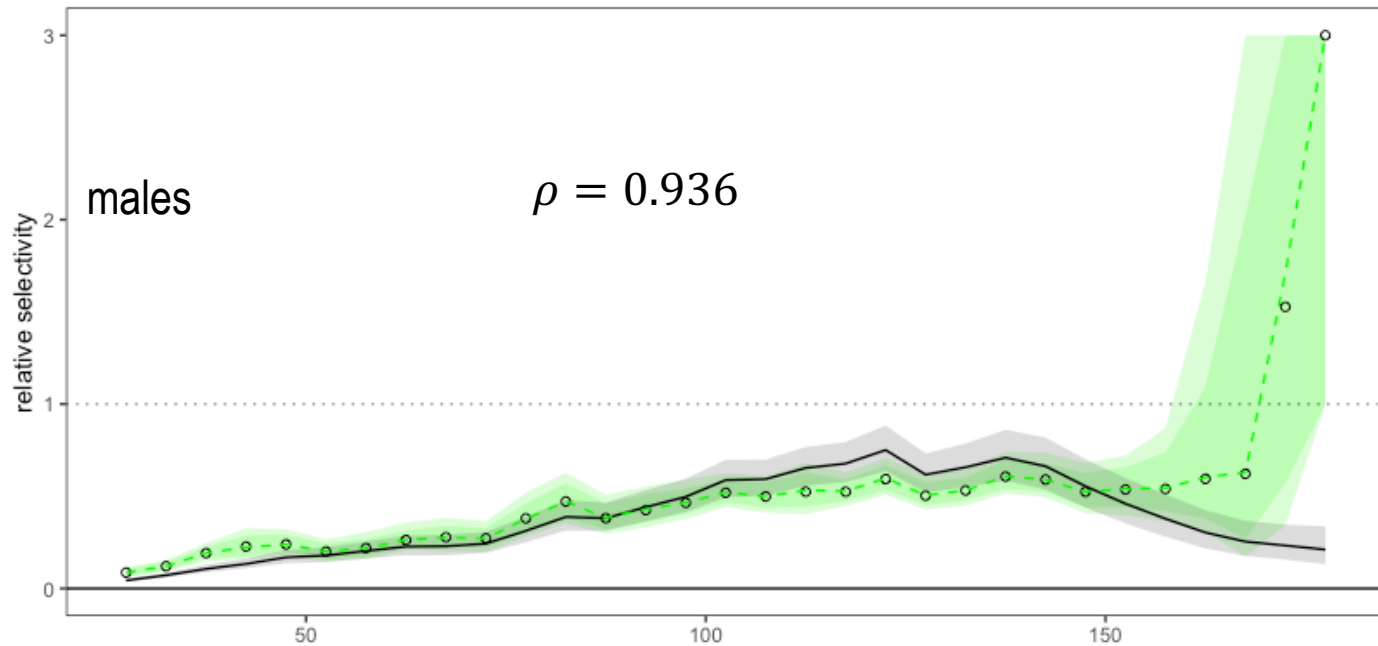
$\Phi_{i(k+1)} - \Phi_{ik} \sim N(0, \sigma_\Phi^2)$ for $k = 1, \dots, m_1 - 1$, In-scale change is RW in size, independent between stations

$$R_{ijk} = R_{ijk}^{\text{WN}} + R_{ijk}^{\text{AR}} \quad R_{ijk}^{\text{WN}} \sim N(0, \sigma_{\text{WN}}^2)$$

$R_{ijk}^{\text{AR}} \sim N(0, \sigma_{\text{AR}}^2)$ $\mathbb{E}(R_{ijk}^{\text{AR}} R_{ijk'}^{\text{AR}}) = \sigma_{\text{AR}}^2 \rho^{|k-k'|}$ In-scale changes are AR1 in size
independent WN between stations, gear

19

- model for joint distribution of catch numbers
- implemented in TMB
- estimates variance-covariance parameters: $\sigma_S^2, \sigma^2, \sigma^2, \sigma^2, \rho$
- many random effects associated with: Φ, R, S ,
- likelihood maximized to estimate var-covar parameters
- posterior modes for S and other random effects R, S ,



- green: bootstrapping estimate and CIs
- black/grey: log-Gaussian Cox model estimate

Summary

- Estimated selectivity curves fairly similar across approaches
- Fully-selected catchability estimates similar across approaches
 - ~0.5 for males
 - ~0.3 for females
- wide range of uncertainty estimates
- Recommendations?